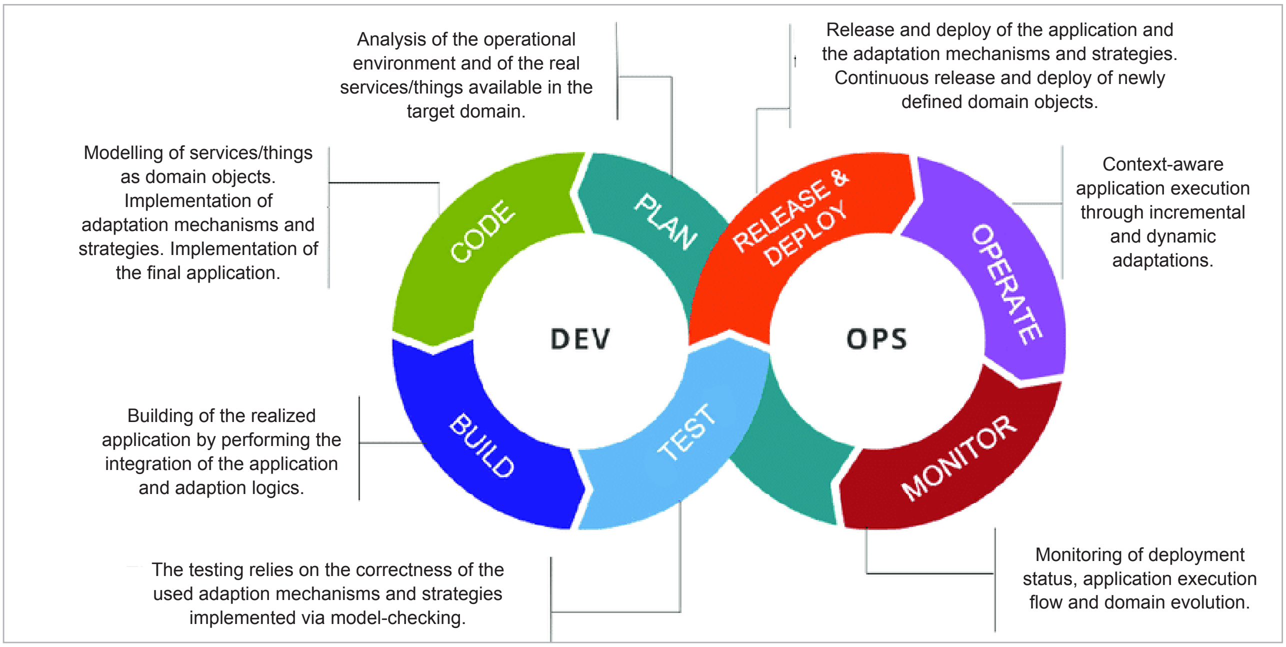

Figure 1: Detailed DevOps workflow

can just start to build on top of these like a matrix. This is a convenience, especially for those who rely heavily on these libraries. A good example of this is what React and Angular have been doing for Web development.

The con: Lagging security practices

However, software security still lags behind in spite of all the progress that is being made. We get to hear about major data breaches occurring around the world and often in large companies even to this day. One of the biggest data breaches happened at Equifax in 2017 on 13th May and was not discovered until 29th July! Once the breach was discovered, effective action was taken within 24 hours. However, a lot of sensitive data such as credit card details, social security numbers, and other important documents were accessed over the two-month period. According to some estimates, almost half of the American population was impacted.

Damages incurred due to poor security

In India, awareness about such security concerns is low, so the severity of this might be hard to grasp for us. These types of data breaches tend to occur from time to time in our country as well. Here are some of the major data breaches that have taken place in India just over the last year.

There was a data breach at Bigbasket in October 2021, which sacrificed the information of over 20 million people. Then there was a breach at Juspay, a payment gateway, in which over 35 million people's primary information such as names and addresses, as well as financial details such as card numbers and banking credentials were revealed. In a security breach at Dominos, the data of over one million people was put at risk. Another major breach occurred very recently at Air India, where over 4.5 million people were affected.

According to research by IBM, it has been found that the average cost of a data breach is over 5 million US dollars. But what is interesting is that it is about 30 times cheaper to fix the defects during the development stage of software, compared to once it goes out into production. Though this is still a considerable amount of money, at least it saves companies from incurring a bigger loss.

So the people behind the development of DevOps are now fostering a new paradigm or a philosophy, which they term as 'Dev + Sec + Ops' (philosophy of integrated security practices within the DevOps process). This basically integrates security practices with the existing DevOps process and philosophy to make sure that security is the most important aspect in every stage of software development, and not an afterthought.

What exactly is DevSecOps?

DevSecOps is still in its early phases and is continuing to mature. There are already a whole bunch of tools available, but there is still plenty of work to be done. The first important aspect that it covers is that it helps in identifying the security issues, unlike in DevOps where these are generally just an afterthought. The security team goes through a higher level of overview and makes sure that security is on point, before giving the final nod in the development process.

DevSecOps also gives speed and agility to the security team so they can